



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2024-25(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

END SEMESTER

Semester: V

Programme: B. Tech Information Technology

Course: Data Warehousing and Mining

Time: 3 Hours

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33505	Course Outcome
CO1	To classify the fundamentals, need, structure, and cost of data warehousing for effective data management.
CO2	To classify data warehousing architecture, modeling techniques, ETL processes, and OLAP operations for efficient data analysis.
CO3	To explain data preprocessing techniques and transformation for improving data quality.
CO4	To provide fundamental knowledge of data mining concepts, techniques, and applications.
CO5	To introduce classification techniques and equip students with the ability to compare and apply different classifiers effectively.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	What is Data Mart?	CO1	2	2
b.	Define Data Cube.	CO2	2	2
c.	What is data cleaning?	CO3	3	2
d.	Define classification in data mining.	CO4	2	2
e.	What is Hadoop?	CO5	2	2

Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain need and advantages of data warehousing.	CO1	2	5
b.	Describe characteristics of data warehouse.	CO1	2	5
c.	Differentiate OLTP and OLAP systems with examples.	CO1	2	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain star schema with neat diagram.	CO2	2	5
b.	Compare star and snowflake schema.	CO2	3	5
c.	Explain multidimensional data model and data cube.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain data preprocessing steps in detail.	CO3	4	5
b.	Describe data integration and its challenges.	CO3	3	5
c.	Explain methods for handling missing and noisy data.	CO3	3	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain different data mining tasks with examples.	CO4	2	5
b.	Describe data mining process (KDD process).	CO4	4	5
c.	Explain classification vs clustering.	CO4	3	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain BI architecture and its components.	CO5	2	5
b.	Discuss Big Data challenges and solutions.	CO5	3	5
c.	Explain Hadoop ecosystem and its components.	CO5	3	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create